

Female Reservation and the Qualifications of Local Elected Officials in India

Gaurav Sood

Working draft

Abstract

Gender quotas expand access to elected office, but their implications for officeholders' qualifications may vary across settings. We compare winners of women-reserved and open seats in rural Bihar, Uttar Pradesh, Rajasthan and Kerala and in Mumbai and Delhi, and combine these estimates with existing schooling studies. The combined schooling difference is -0.45 standard deviations (95% CI -0.76 to -0.14), with substantial variation across settings. In our records, reserved-seat village heads in Bihar and Uttar Pradesh are 7 to 15 percentage points less likely to hold a university degree. Estimates for Kerala's village ward members and the two cities are smaller and their confidence intervals include zero. Reserved-seat winners are 22 to 55 points more likely to report no occupation in the village, block and city offices with occupation records, and declare lower assets in Rajasthan and Uttar Pradesh. In Mumbai and Delhi, they are 11 to 27 points less likely to face pending criminal charges. Qualification differences between reserved and open seats depend on both the attribute and the office.

1 Introduction

Gender quotas broaden access to elected office. Whether they also change officeholders' qualifications matters for how representative government works. Local officials must interpret administrative rules, oversee expenditure, and negotiate with other officials (Auerbach et al. 2025). Schooling and political experience may help them perform these tasks; economic resources may affect their independence. Yet these attributes need not move together, and none alone establishes how well an official governs. The question is which qualifications change when office is reserved for women.

Research on Indian village heads often finds that winners of women-reserved seats have less schooling and prior political experience than winners of open seats (Chattopadhyay and Duflo 2001; Ban and Rao 2007; Afridi et al. 2013). That evidence has an important qualification. Women who enter politics can be positively selected relative to other women even when they have less schooling than male politicians. Evidence from Karnataka, for example, finds stronger selection on cognitive skills among female than male candidates relative to voters of their own gender (Mori et al. 2025). The qualifications of elected officials depend both on the eligible population and on who enters and wins elections.

This leaves a puzzle. Is the schooling gap a common consequence of reservation, or does it depend on the opportunities and barriers women face in a particular political setting? Much of the existing evidence concerns village heads. It does not establish whether the same pattern holds among other rural officials or urban councillors. Nor does a schooling difference establish a broader qualification disadvantage: economic position and criminal records could follow different patterns.

We extend this literature by comparing qualifications across rural and urban offices in several Indian states. A common university-degree threshold makes the education comparisons easier to interpret across settings. Occupation, assets, age, and criminal records allow us to ask whether the education pattern extends to other attributes. We also combine the existing schooling evidence with our estimates in a meta-analysis, expressing differences in standard-deviation units and counting overlapping samples once. The contribution is to establish how qualification differences vary across offices, settings, and attributes, rather than to infer a single ranking of leader quality.

Our analysis uses election records and candidate affidavits for rural councils in Bihar, Uttar Pradesh, Rajasthan, and Kerala and municipal councils in Mumbai and Delhi. We compare winners of women-reserved and open seats separately by state, election, and office; open seats can also elect women. The regressions include geographic and caste-reservation effects where the records permit them, with standard errors clustered geographically. A causal interpretation requires reservation to be as good as randomly assigned within the comparison groups. We also examine how missing qualifications limit conclusions about all winners.

The education results differ across settings. Among village heads in Bihar and Uttar Pradesh, reserved-seat winners are 7 to 15 percentage points less likely to hold a university degree. Rajasthan also has a negative estimate among winners with classifiable education, but extensive missingness prevents an all-winner conclusion. Kerala’s village ward members and the councillors of Mumbai and Delhi show no consistent graduate-share deficit, and their confidence intervals include zero. The urban intervals are too wide to establish equivalence. Comparing village ward members in Bihar and Kerala preserves the contrast, so it is not solely a difference between heads and members. Differences in women’s education and access to political life are possible explanations, but offices, election years, and data coverage also vary across settings. Our design cannot separate those explanations.

Other attributes follow different patterns. Across the village, block, and city offices with occupation records, reserved-seat winners are 22 to 55 points more likely to report no occupation, including where the estimated education differences are small. They also declare fewer assets in Rajasthan and Uttar Pradesh. In Mumbai and Delhi, however, they are 11 to 27 points less likely to face a pending criminal case. The lower prevalence of charges is a potentially favorable difference, although charges are not convictions and our records do not establish its consequences for governing performance. Reservation is associated with different changes in different qualifications; the schooling evidence alone does not describe that pattern.

2 Related literature

2.1 Evidence on officials’ qualifications

The most consistent evidence on qualifications concerns schooling among village heads. Surveys generally find less schooling among winners of women-reserved seats. The comparisons assembled in Table D.1 report lower years of schooling in West Bengal and South India (Chattopadhyay and Duflo 2001, Table 3; Beaman et al. 2009, Table II; Ban and Rao 2007, Table 9), lower secondary-school attainment in a national village panel and Andhra Pradesh (Deininger et al. 2011, Table 3; Afridi et al. 2013, Table 2). These studies combine descriptive and adjusted comparisons and measure different schooling outcomes. We synthesize differences in qualifications, without treating the estimates as a common causal effect.

The schooling measures range from years of education to completion of secondary school. We

express these comparisons and our graduate-share estimates in standard-deviation units and present the combined meta-analysis in Section 4.4. Not every relevant study supplies an eligible comparison. In Tamil Nadu, female and male presidents have relatively similar reported schooling distributions (Gajwani and Zhang 2015, 246). In Maharashtra, female sarpanches have fewer years of schooling, but the study compares women and men, including a woman elected without a current reservation, rather than reserved and open seats (Priebe 2017, Table 2 and note 6).

Schooling is one component of preparation for office. Reserved-seat leaders also have less prior political experience in West Bengal, South India, and Andhra Pradesh (Chattopadhyay and Dufflo 2001, Table 3; Ban and Rao 2007, Table 9; Afridi et al. 2013, Table 2). Local officials must also understand administrative procedures, obtain information, and negotiate with other officials. Village heads’ responsibilities for budgets, public works, and benefit allocation make these skills relevant (Auerbach et al. 2025, 708–10; Joshi et al. 2025, 56). In Tamil Nadu, female presidents score 3.2 points lower on a 19-question test of panchayat rules and duties after adjustment for schooling and other observed characteristics. They also report less contact with higher-level officials (Gajwani and Zhang 2015, Tables 5–6). The knowledge coefficient is an adjusted gender comparison, not an estimate of the causal contribution of schooling. It nevertheless shows why similar educational attainment need not imply similar preparation for governing.

2.2 Possible mechanisms

Lower schooling relative to open-seat winners can coexist with positive selection relative to other women. In South India, reserved-seat leaders are wealthier and more politically informed than eligible women (Ban and Rao 2007, Table 9). Evidence from Karnataka similarly finds that female candidates are more positively selected on cognitive skills relative to female voters than male candidates are relative to male voters (Mori et al. 2025). For Bihar’s 2016 and 2021 elections, Bamezai et al. (2026) report lower schooling among reserved-seat winners near reservation thresholds, using local citizens as the benchmark. We discuss that evidence separately because it measures a local effect on citizen-standardized schooling, not the broad winner comparison synthesized here. Differences between eligible populations and differences in whom elections draw from each population are distinct potential sources of qualification gaps. Our winner records cannot separate them.

Bhavnani, Huidobro and Prillaman’s October 2024 draft also reports lower schooling in women-reserved seats. For Odisha’s 2022 election, the adjusted difference is -1.08 years among seats without SC/ST reservation (Bhavnani et al. 2024, Table 2, column 4). Their claims about equal or higher quality concern household-wealth residuals and education relative to one’s identity group. Those measures address different questions from attained schooling and do not directly measure governing performance. The appendix reconstructs the caste-specific schooling comparisons.

Gender quotas can also change representation along dimensions other than gender. In the Rural Economic and Demographic Survey, reservation for women reduces high-caste candidacy and representation; differences in women’s mobility across caste groups provide a possible explanation (Cassan and Vandewalle 2021, Tables 1–2 and 9–11). Education, assets, caste, and family political experience may therefore change together. Our controls for a seat’s caste-reservation category distinguish institutional categories, but do not isolate the mechanisms through which candidates enter or voters choose among them.

Experience can alter these differences after entry into office. In Andhra Pradesh, councils reserved for women initially show more leakage and implementation problems in a public employment program, but audit outcomes improve as female leaders accumulate experience (Afridi et al. 2013).

Evidence from South India also links political experience to women’s performance in office (Ban and Rao 2007). Among citizens in Uttar Pradesh, gaps in political participation persist after accounting for education, political knowledge, assets, and restrictions on mobility, suggesting that observed qualifications do not exhaust the barriers to participation (Iyer and Mani 2019). Entry qualifications and the opportunities to acquire governing experience are distinct objects of study.

Formal office also differs from authority over decisions. Research on rural councils documents limits on women’s participation in deliberation and the involvement of male relatives in everyday governance (Heinze et al. 2025; Amar et al. 2025; Karekurve-Ramachandra and Sood 2026). Publicly reported work on collective family governance finds family involvement in both male- and female-led councils, alongside lower political authority for women (Huidobro et al., n.d.). These findings caution against interpreting every family connection as evidence of substitution for the elected official. Our occupation and asset measures describe economic position; they do not directly measure control over household resources or the exercise of political authority. A declaration of no occupation may also encompass substantial unpaid work.

2.3 Qualifications and governing performance

The association between formal qualifications and performance is itself an empirical question. More educated council heads in South India allocate benefits differently, including greater targeting to landless households (Besley et al. 2007, Tables 4–5). Graduates who narrowly win Indian state assembly elections increase constituency night-time-light growth (Jain et al. 2022, Table 2). Evidence from national leaders, US legislators, and Brazilian mayors, however, does not establish a general performance advantage for college graduates (Carnes and Lupu 2016). Other attributes have their own implications: narrowly electing legislators who face criminal charges reduces growth and road provision in India (Prakash et al. 2019), while younger Italian mayors facing reelection exhibit larger electoral budget cycles (Alesina et al. 2019, Tables 5–7). None of these findings warrants combining schooling, age, wealth, and criminal records into a single ranking of leader quality.

The effects of reservation on governance extend beyond the credentials of the elected official. In West Bengal and Rajasthan, reservation changes public investment toward goods requested by women (Chattopadhyay and Duflo 2004, Table V). Within India, the evidence varies by outcome and setting. Tamil Nadu’s female-led councils build fewer schools and roads (Gajwani and Zhang 2015, Tables 7–8), while a broader village survey finds greater public-goods availability and fewer reported bribes despite lower satisfaction with female leaders (Duflo and Topalova 2004, Tables 2–4). In Maharashtra, improvements in women’s political participation coexist with mixed service-delivery differences (Priebe 2017, Tables 4 and 6). Repeated exposure to female leaders in West Bengal also raises girls’ aspirations and educational attainment (Beaman et al. 2012, Tables 1–3). These outcomes concern policy, participation, and longer-run opportunities, which our records of officeholders do not measure.

The unresolved question for this paper is how broadly the qualification differences documented among village heads extend across offices and attributes. We address that question with comparable measures where the administrative records permit them, while retaining separate estimates for each setting.

3 Institutional setting, data, and empirical strategy

3.1 Reserved seats and the comparison

Women-reserved seats restrict eligibility to women; open seats permit both women and men to contest. Gender reservation also operates within caste-reservation categories. These rules change eligibility, but their assignment procedures differ across jurisdictions. We therefore retain separate state, election, and office comparisons and make the identifying assumption explicit below. Differences across settings also reflect differences in offices, election years, and source coverage; they do not by themselves identify an effect of a state’s social or institutional environment.

3.2 Data sources

The administrative records come from state election repositories through `local_elections`, with input hashes, winner rules, exclusions and coding dictionaries saved in the repository. Bihar’s winner is the highest vote-getter, an uncontested candidate, or the winner of a tie drawn by lot; seats whose results form cannot identify a winner are excluded. Kerala’s records are of elected ward members, and their reservation refers to the ward even for members who later lead the council.

Bihar’s 2021 records include age but not education. We obtain reservation status from the State Election Commission’s feed for the 2021–2026 term and verify that it reproduces the Commission’s count of women-reserved village-head seats. Kerala records education as free text; we classify recognized degrees and school qualifications and leave incomplete or unrecognized entries missing.

Mumbai’s councillor records come from the Praja Foundation’s report cards and affidavits, as deposited by Karekurve-Ramachandra and Lee (2023). Each councillor with a recorded age enters once per term.

The Delhi data cover the municipal elections of 2012, 2017, and 2022. Winners and assembly mapping for 2012 and 2017 come from Goyal’s replication deposit (Goyal 2023); for 2022, winners come from the OpenCity compilation of Lok Dhaba results and reservation from the State Election Commission’s gazette (Vaidyanathan 2023; State Election Commission, NCT of Delhi 2022, Annexure B). The 2012 reservation labels match the official schedule (State Election Commission, NCT of Delhi 2012, Annexures I–III). Qualifications and pending cases come from each winner’s MyNeta profile (Association for Democratic Reforms 2026); convictions are excluded from pending cases. We use these profiles to measure Delhi winners’ education because the older deposited classifications disagree with the affidavits. The measurement appendix reports that comparison.

3.3 Outcomes and samples

The main education outcome is whether the winner holds a university degree; the appendix also reports illiteracy. Unknown qualifications stay missing. Occupation is recorded in Rajasthan, Kerala and Delhi. A winner counts as having no occupation when the entry is nil, unemployed, homemaker or student; pensioners and every named trade or job count as an occupation, with the coding rules recorded in the accompanying analysis code.

Declared assets come from the nomination affidavits of Rajasthan’s 2020 and Uttar Pradesh’s 2021 village heads. We analyze declared assets on a logarithmic scale.

MyNeta’s affidavit summaries record pending criminal cases in Mumbai and Delhi. Uttar Pradesh’s 2021 affidavits instead provide an indicator of reported criminal history.

3.4 Specification and inference

For each state, election and office, we estimate

$$Y_i = \alpha + \beta R_i + \gamma_{g(i)} + \delta_{c(i)} + \epsilon_i,$$

where R_i indicates a seat reserved for women, $\gamma_{g(i)}$ is a block effect (a district effect where blocks are not recorded throughout) and $\delta_{c(i)}$ is an effect for the seat’s caste-reservation class. For binary outcomes, 100β is the reserved-minus-open difference in percentage points among seats in the same block and caste class. It is the effect of reservation if reservation is as good as randomly assigned within those cells. Standard errors are clustered by block or district, and 95% intervals use a t distribution with one fewer degree of freedom than there are clusters; Kerala’s ward and district-member models have only 14 clusters. Mumbai pools its 2012 and 2017 councils with administrative-ward and council effects, clustered by electoral ward, and has no caste-reservation control. Delhi uses assembly-constituency and caste-reservation effects in each election, clustered by assembly constituency.

3.5 Combining the schooling evidence

The meta-analysis combines existing reserved-versus-open schooling comparisons with this paper’s estimates. Years of schooling are expressed relative to their within-sample standard deviation. For binary schooling thresholds, we convert log odds ratios to the same approximate scale. Our threshold estimates use logistic regressions with the controls and clustering described above. This conversion assumes that the thresholds measure a common underlying schooling distribution; it does not make secondary completion and a university degree the same qualification.

We fit a random-effects model that allows differences across studies and settings, with repeated elections nested within each setting. The pooled estimate describes the assembled evidence, not an average over all Indian local offices. We retain one Birbhum study and exclude citizen-relative selection measures and local RD estimates from the pool. The appendix gives the conversion, model coverage, and sensitivity results.

4 Results

4.1 Rural panchayats

4.1.1 Village heads

Among village heads in Bihar and Uttar Pradesh, winners of women-reserved seats are less likely to hold a university degree. In Bihar’s 2016 election, the adjusted graduate-share difference is -15.0 percentage points (95% CI -16.5 to -13.4). Uttar Pradesh shows the same direction in each election. Its estimated differences are -13.1 percentage points (95% CI -14.5 to -11.7) in 2010, -9.1 percentage points (95% CI -9.7 to -8.5) in 2015, and -7.4 percentage points (95% CI -8.1 to -6.7) in 2021. The estimated deficit narrows over time, but all three intervals remain below zero. These are comparisons of the winners in each election, not changes in the qualifications of the same individuals.

Rural village heads: graduate or above

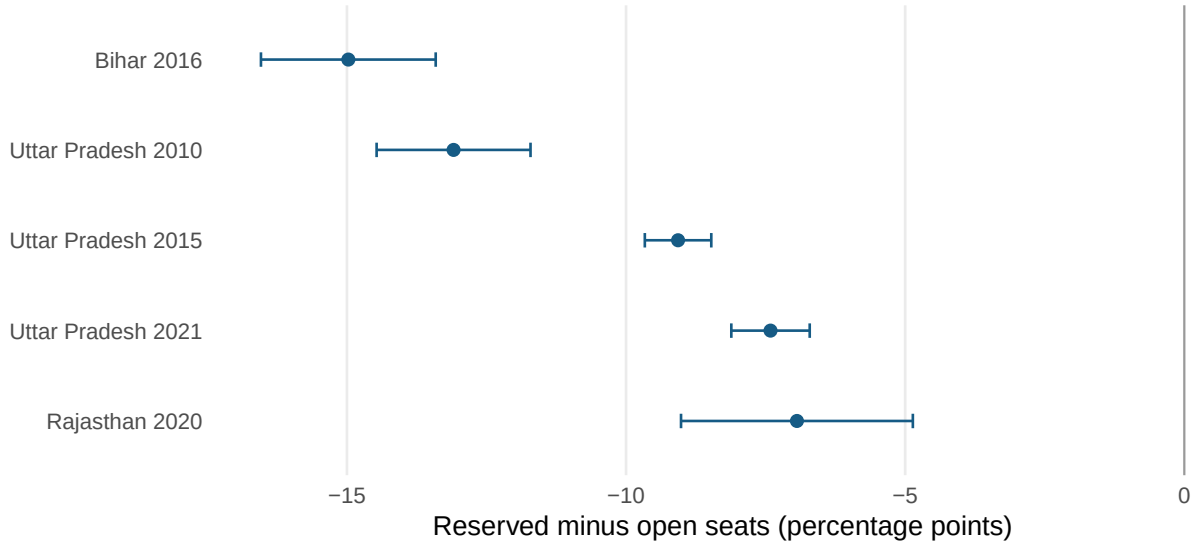


Figure 1: Graduate-share differences among rural village heads, women-reserved minus open seats, with geography and caste-reservation effects. Bars are 95% confidence intervals clustered by geography. UP 2010 uses districts; other rows use blocks. Unclassified education, ambiguous winners and unknown reservation are excluded.

Rajasthan also has a negative education difference among village heads whose qualifications can be classified: -6.9 percentage points (95% CI -9.0 to -4.9). Here, however, missing education changes the conclusion we can draw. Many records report only “Other,” and the share with classifiable education differs between reserved and open seats. Allowing each unclassified winner to be either a graduate or a nongraduate yields an all-winner coefficient between -29.7 and 31.5 percentage points. This range allows substantial differences in either direction. The observed negative gap therefore describes the classified sample; it does not establish a graduate-share deficit among all Rajasthan village heads. These bounds address missing outcomes and do not incorporate sampling uncertainty.

4.1.2 Bihar’s other offices

The education difference in Bihar extends beyond village heads. The 2016 records cover 210,717 elected seats across six village, block, and district offices. Reserved-seat winners have lower graduate shares in every office, although the size of the difference varies. Among village ward members, the adjusted difference is -4.6 percentage points (95% CI -4.9 to -4.2); among block council members it is -10.8 percentage points (95% CI -12.0 to -9.5). The gap is thus present among council members as well as heads, rather than being confined to the village executive office.

Bihar 2016: graduate or above

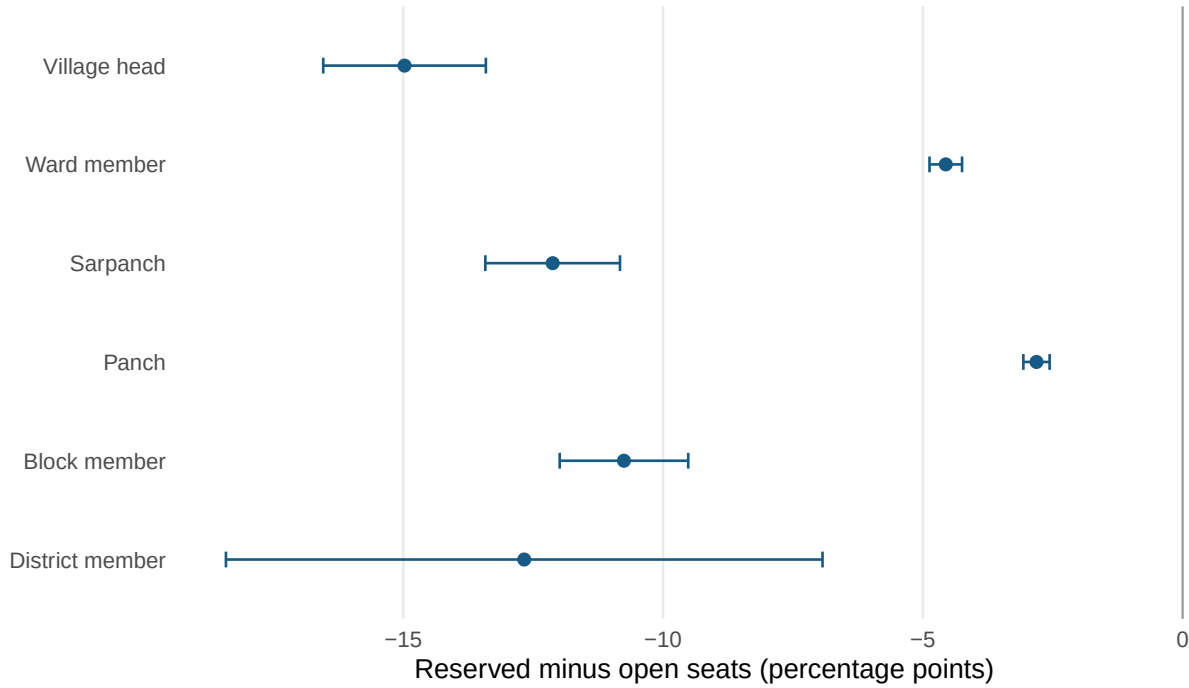


Figure 2: Bihar 2016: graduate-share differences by office, women-reserved minus open seats, with block and caste-reservation effects (district effects for district council members). Bars are 95% confidence intervals clustered at that level.

Reserved-seat village heads are also younger. The estimated age difference is 3.6 years in 2016 and 3.2 years in 2021 (95% CI 2.7 to 3.6). The 2021 data cover 244,474 elected seats but contain no education measure, so they cannot establish whether the schooling gap persisted. Age describes another difference in who holds office; without records of prior political experience, it cannot establish that younger winners are less prepared to govern.

4.1.3 Kerala

Kerala's village ward members show much smaller education differences. The adjusted graduate-share differences are -0.5 percentage points (95% CI -2.4 to 1.3) in 2010, -0.1 percentage points (95% CI -1.6 to 1.3) in 2015, and 2.0 percentage points (95% CI -0.8 to 4.7) in 2020. Each interval includes zero, and none includes a deficit as large as the one estimated for Bihar's village ward members. Comparing this common office therefore preserves the contrast between Bihar and Kerala; the pattern is not simply a consequence of comparing northern heads with Kerala's council members.

Kerala: graduate or above

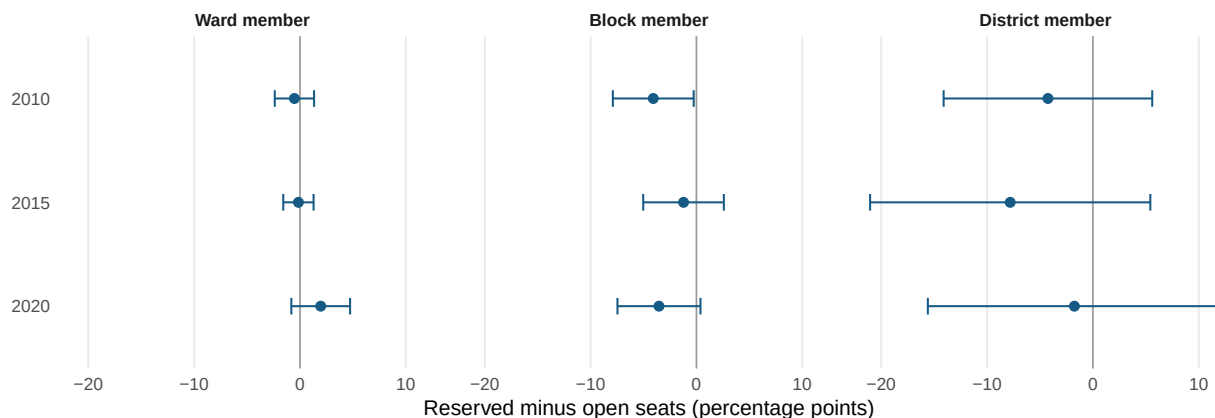


Figure 3: Rural Kerala: graduate-share differences by election and office, women-reserved minus open seats, with caste-reservation effects. Block members use block effects and clustering; village and district members use districts. Bars are 95% confidence intervals.

This result applies most clearly to village ward members, not to every Kerala office. For block council members, the estimated differences are negative in all three elections, and the 2010 interval excludes zero (-4.1 percentage points (95% CI -7.9 to -0.3)). The district council estimates are also negative but substantially less precise. Moreover, these comparisons describe winners with classifiable education. The missing-outcome bounds for Kerala permit either sign, so small estimated gaps in the observed sample do not establish equal educational attainment among all elected officials.

4.2 Urban councils

4.2.1 Mumbai

Mumbai does not show the graduate-share deficit found among northern village heads. Across the 2012 and 2017 councils, the adjusted difference is 8.4 percentage points (95% CI -0.4 to 17.3). The point estimate favors reserved-seat winners, but its interval includes zero. Within the observed sample and regression specification, the interval permits a negligible disadvantage or a substantial advantage; it provides little support for a deficit of the magnitude observed among northern village heads.

Urban: Mumbai councillors, 2012 and 2017

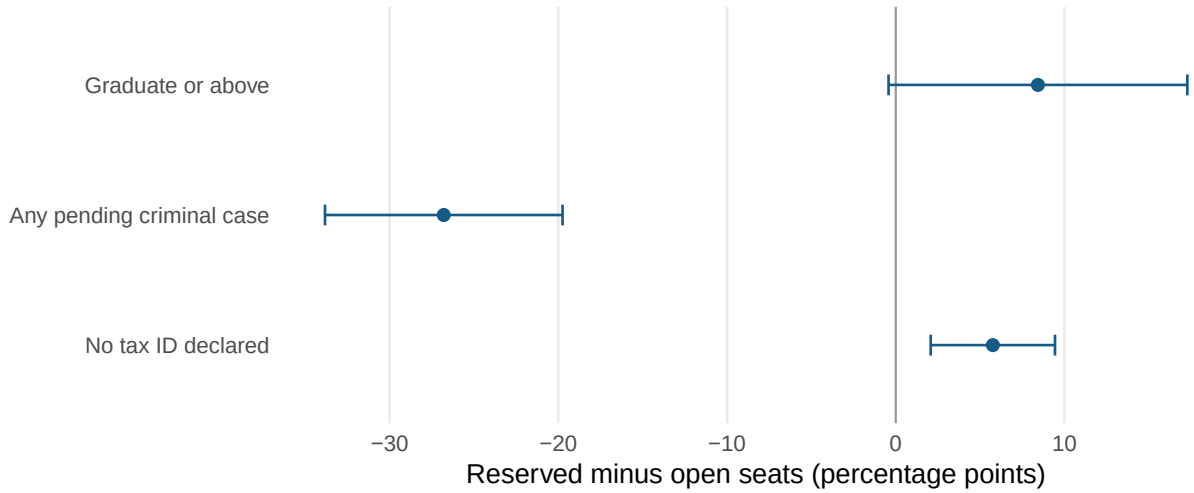


Figure 4: Mumbai councillors in the 2012 and 2017 councils: women-reserved minus open seats in graduate share, any pending case and declaring no tax ID (PAN), with administrative-ward and council effects. Bars are 95% confidence intervals clustered by electoral ward.

The difference in pending criminal cases is larger and more precisely estimated. Reserved-seat councillors are 26.8 percentage points less likely to face a pending case (95% CI 19.7 to 33.8). This is a difference in the prevalence of charges against elected officials. It does not measure convictions or establish how reservation affects misconduct or performance in office.

4.2.2 Delhi

In Delhi, the estimated graduate-share differences are small, but their uncertainty is substantial. The estimated graduate-share differences are -0.6 percentage points (95% CI -14.4 to 13.2) in 2012, 1.1 percentage points (95% CI -10.4 to 12.6) in 2017, and -3.4 percentage points (95% CI -15.9 to 9.1) in 2022. The estimates vary in sign, and all three intervals include sizable differences in both directions. The Delhi results therefore do not establish an education advantage or disadvantage, and they are too imprecise to establish equivalent graduate shares.

Urban: Delhi councillors

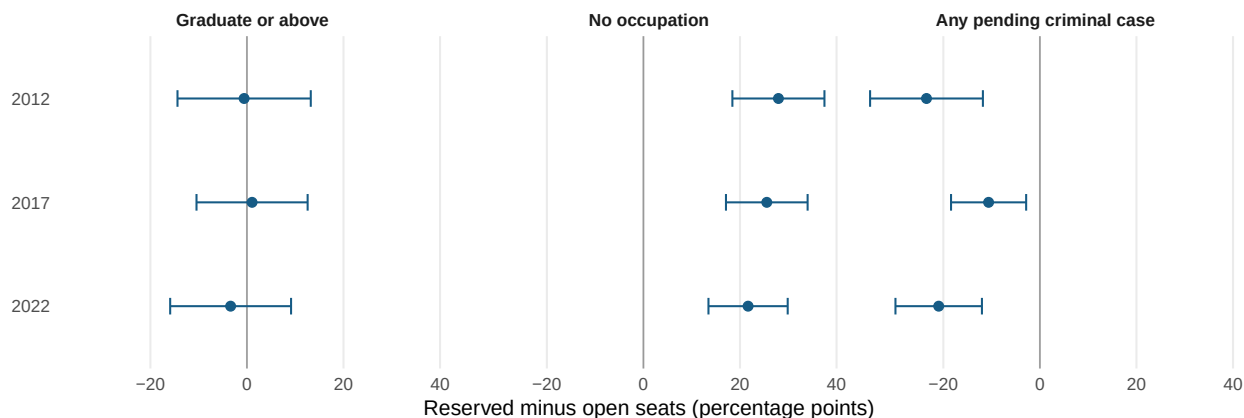


Figure 5: Delhi councillors by election: women-reserved minus open seats in graduate share, reporting no occupation and any pending case, with assembly-constituency and caste-reservation effects. Bars are 95% confidence intervals clustered by assembly constituency.

Pending criminal cases show a more consistent pattern. The reserved-minus-open differences are -23.5 percentage points (95% CI -35.2 to -11.8) in 2012, -10.6 percentage points (95% CI -18.4 to -2.8) in 2017, and -21.0 percentage points (95% CI -29.9 to -12.0) in 2022. All three intervals lie below zero. The sign also remains negative when every missing case indicator is allowed to be either zero or one, holding the regression design fixed. These bounds address incomplete records and do not incorporate sampling uncertainty. They also cannot establish the accuracy of declarations or the consequences of pending cases for governance. The urban evidence is consequently clearer about the lower prevalence of charges than about differences in educational attainment.

4.3 Occupation, assets and criminal records

Differences in reported occupation extend to settings where the education estimates are close to zero. In Rajasthan, the share reporting no occupation is 51 percentage points higher among reserved-seat village heads (95% CI 48 to 54). Among Kerala’s village ward members, the estimated differences range from 41 to 55 points across the three elections; among Delhi’s councillors, they range from 22 to 28 points. Thus, the small education differences in these settings coexist with large differences in reported occupation. Kerala’s district council members show smaller occupation differences, ranging from 0 to 15 points.

The occupation measure combines declarations of no work, unemployment, homemaking, and student status. It describes reported occupation, not the absence of productive activity or access to household resources. Coding “social worker” as missing rather than as an occupation changes no estimate by more than 6 percentage points, so that particular coding choice does not account for the broad pattern. The records do not show whether women with no reported occupation control assets or exercise political authority independently.

Mumbai’s records contain no occupation measure. The adjusted difference in the share declaring no tax identification number (PAN) is 5.8 percentage points (95% CI 2.1 to 9.4), with nondeclaration more common among reserved-seat winners. Nondeclaration occurs only in the 2012 council records. This is an additional difference in affidavit disclosure; it cannot substitute for a direct measure of employment or income.

Reserved-seat village heads declare fewer assets in both states with asset records. The estimates imply approximately 47% lower assets in Rajasthan (95% CI 40% to 54% lower) and 9% lower assets in Uttar Pradesh (95% CI 5% to 13% lower). The asset differences reinforce the evidence from occupation that reserved-seat winners occupy a different economic position. Affidavits record declared holdings, however, and do not establish who controls those resources.

The urban difference in pending cases does not have a clear counterpart in Uttar Pradesh’s village-head records. Only 0.18% of winners with a known record report a criminal history. The adjusted reserved-minus-open difference is -0.04 percentage points (95% CI -0.12 to 0.04), leaving the direction uncertain. This outcome is reported criminal history, whereas the urban measure concerns pending cases, so differences across these settings also reflect the measures available. The appendix reports the full estimates and sample sizes.

4.4 Schooling across the literature and our data

Combining the existing literature with our data, winners of women-reserved seats have less schooling on average, but the size of the difference varies substantially. The pooled difference is -0.45 SD (95% CI -0.76 to -0.14), across 16 estimates from 4 literature samples and 6 settings in our data. Figure 6 shows where that average comes from. The published village-head comparisons and our northern village-head estimates are negative throughout. Our Kerala ward-member and city estimates are much closer to zero. The model’s 95% prediction interval, -1.45 to 0.55 SD, reflects that variation: the negative average does not imply a schooling deficit in every setting.

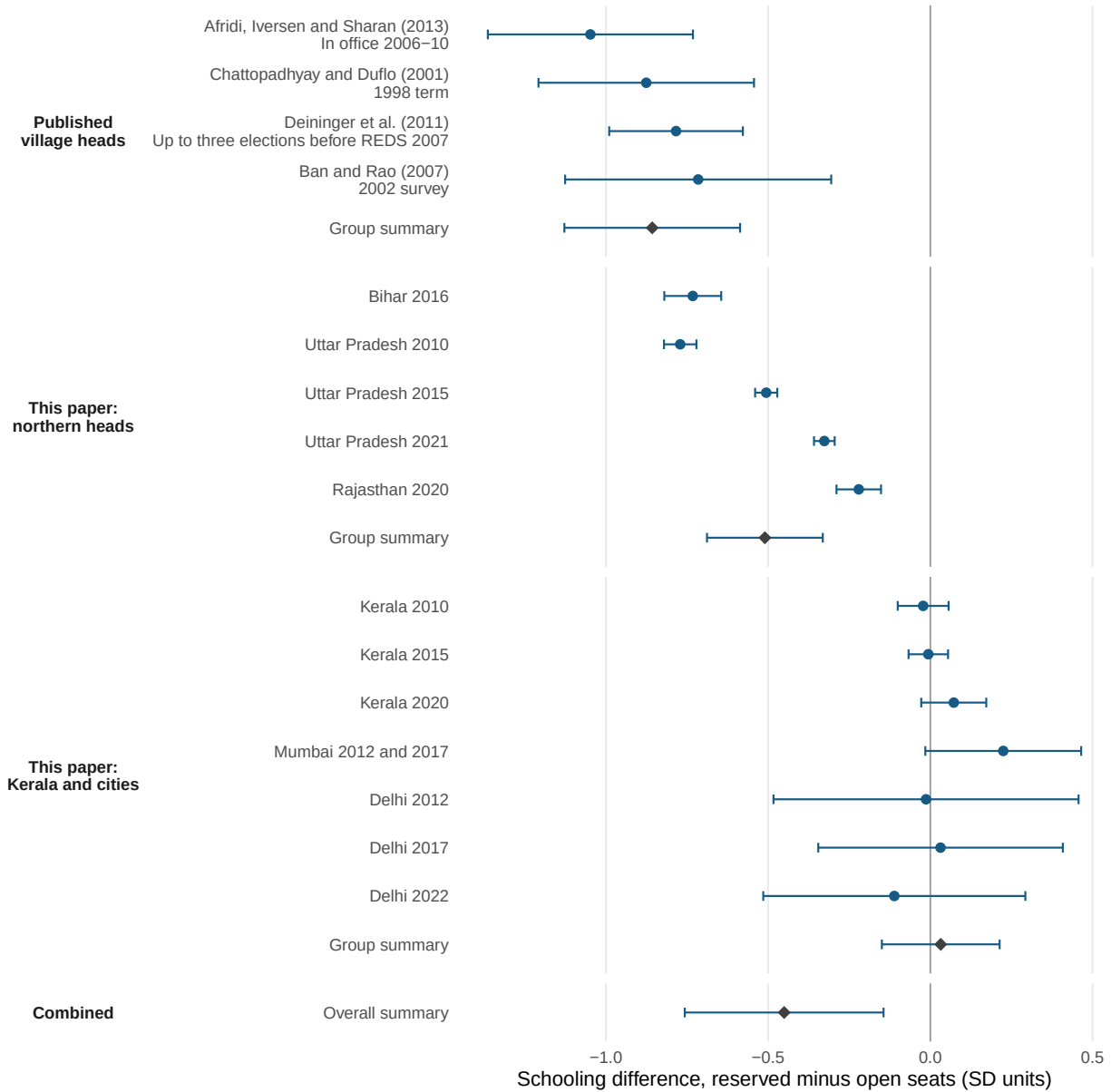


Figure 6: Schooling differences, women-reserved minus open seats, combining the literature and this paper. Circles are individual estimates and diamonds are pooled means; bars show 95% confidence intervals. Years of schooling use pooled within-group SDs; binary thresholds use a log-odds conversion. Our estimates retain geography and reservation controls and clustered uncertainty. Group means come from a model with source/setting-group indicators; the overall mean comes from a model without them. Both allow elections within settings. Birbhum enters once; citizen-standardized RD estimates are excluded. Appendix A gives conversion assumptions and sensitivity checks.

The graduate-share deficit is larger among northern village heads than among Kerala’s village ward members and the two cities’ councillors. Pooling this paper’s estimates with elections nested in settings, reserved-seat village heads in the northern states are 10.5 points less likely to hold a degree (95% CI 5.2 to 15.8), against a difference of 1.4 points (-4.9 to 7.6) among Kerala’s ward members and the two cities’ councillors. Office, data source and election years vary along with setting, and

the second group is one state and two cities. Replacing Bihar’s village heads with its ward members, the office we observe in Kerala, still leaves a northern gap of 7.5 points.

The combined deficit is not driven by one study or setting. Omitting each study or setting in turn gives pooled differences from -0.52 to -0.39 SD; every interval remains below zero. Replacing the Birbhum comparison, excluding Rajasthan, and allowing correlated sampling errors across elections also preserves the negative pooled difference (Table A.2). These checks support the average pattern in the assembled evidence. They do not resolve differences in schooling measures, research designs, or the coverage of winners with recorded education.

5 Discussion

The schooling deficit documented among village heads does not extend uniformly to other local offices. Our evidence from Kerala’s ward members and the two cities places a limit on that generalization, while the combined meta-analysis establishes a negative average across the available comparisons. The contrast is consistent with differences in women’s educational opportunities and selection into politics. Because location, office, election year and measurement vary together, we cannot attribute it to any one mechanism.

Where the education estimates are close to zero, the differences in reported occupation remain large. This pattern is consistent with unequal economic opportunities, but occupation records alone cannot establish who controls resources or exercises the office. In a Rajasthan phone audit, 9% of answered calls to reserved-seat village heads reached the elected woman, against 76% in open seats and 31.7% in urban Jaipur (Karekurve-Ramachandra and Sood 2026, 1–2). Women elected through quotas participate less in council decisions (Heinze et al. 2025), and surveys in two states find male family members prominent in everyday governance (Amar et al. 2025). For the same reason, the lower rate of pending cases among urban reserved-seat councillors may overstate the change in the office’s criminal exposure: affidavits list the officeholder’s own cases, not her relatives’, and Karekurve-Ramachandra and Lee (2020a, 11) note that many Mumbai quota women have male relatives who face charges.

Assessing how qualification differences affect governing performance requires direct measures of officials’ behavior. Separate evidence from Rajasthan and Uttar Pradesh finds no detectable effect of lottery-assigned reservation on the scale or content of public works (Green et al. 2026). Our administrative records establish differences in who holds office; they cannot establish the consequences of those differences.

References

- Afridi, Farzana, Vegard Iversen, and M. R. Sharan. 2013. *Women Political Leaders, Corruption and Learning: Evidence from a Large Public Program in India*. No. 7212. IZA. <https://docs.iza.org/dp7212.pdf>.
- Alesina, Alberto, Traviss Cassidy, and Ugo Troiano. 2019. “Old and Young Politicians.” *Economica* 86 (344): 689–727. <https://doi.org/10.1111/ecca.12287>.
- Amar, Priyadarshi, Apurva Bamezai, and Rithika Kumar. 2025. “Missing from the Middle: Patriarchal Constraints and the Invisibilization of Women in Everyday Governance.” <https://doi.org/10.2139/ssrn.5200785>.
- Association for Democratic Reforms. 2026. *Delhi Municipal Election Winner Profiles: 2012, 2017 and 2022*. <https://www.myneta.info/>.
- Auerbach, Adam Michael, Shikhar Singh, and Tariq Thachil. 2025. “Who Knows How to Govern? Procedural Knowledge in India’s Small-Town Councils.” *American Political Science Review* 119 (2): 708–26. <https://doi.org/10.1017/S0003055424000297>.
- Bamezai, Apurva, Siddharth George, M. R. Sharan, and Borui Sun. 2026. “The Determinants and Consequences of Political Selection in the Developing World: Evidence from India.” <https://www.mrsharan.com/papers/political-selection.pdf>.
- Ban, Radu, and Vijayendra Rao. 2007. “Tokenism or Agency? The Impact of Women’s Reservations on Village Democracies in South India.” https://vijayendraraio.org/papers/EDCC_Reservations_to_go.pdf.
- Beaman, Lori, Raghabendra Chattopadhyay, Esther Duflo, Rohini Pande, and Petia Topalova. 2009. “Powerful Women: Does Exposure Reduce Bias?” *Quarterly Journal of Economics* 124 (4): 1497–540. <https://www.povertyactionlab.org/sites/default/files/research-paper/310%20Female%20Quotas%20Nov%2009.pdf>.
- Beaman, Lori, Esther Duflo, Rohini Pande, and Petia Topalova. 2012. “Female Leadership Raises Aspirations and Educational Attainment for Girls: A Policy Experiment in India.” *Science* 335 (6068): 582–86. <https://doi.org/10.1126/science.1212382>.
- Besley, Timothy, Rohini Pande, and Vijayendra Rao. 2007. “Just Rewards? Local Politics and Public Resource Allocation in South India.” https://vijayendraraio.org/papers/justrewards_oct07.pdf.
- Bhavnani, Rikhil R., Alba Huidobro, and Soledad Artiz Prillaman. 2024. “Does Affirmative Action Worsen Quality? Theory and Evidence to the Contrary from Elections.” <https://rbhavnani.github.io/files/BHPQuotasQuality.pdf>.
- Carnes, Nicholas, and Noam Lupu. 2016. “What Good Is a College Degree? Education and Leader Quality Reconsidered.” *The Journal of Politics* 78 (1): 35–49. <https://doi.org/10.1086/683027>.
- Cassan, Guilhem, and Lore Vandewalle. 2021. “Identities and Public Policies: Unexpected Effects of Political Reservations for Women in India.” *World Development* 143: 105408. <https://doi.org/10.1016/j.worlddev.2021.105408>.
- Chattopadhyay, Raghabendra, and Esther Duflo. 2001. *Women as Policy Makers: Evidence from a India-Wide Randomized Policy Experiment*. No. 8615. National Bureau of Economic Research. https://www.nber.org/system/files/working_papers/w8615/w8615.pdf.

- Chattopadhyay, Raghabendra, and Esther Duflo. 2004. "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India." *Econometrica* 72 (5): 1409–43. <https://doi.org/10.1111/j.1468-0262.2004.00539.x>.
- Chinn, Susan. 2000. "A Simple Method for Converting an Odds Ratio to Effect Size for Use in Meta-Analysis." *Statistics in Medicine* 19 (22): 3127–31. [https://doi.org/10.1002/1097-0258\(20001130\)19:22%3C3127::AID-SIM784%3E3.0.CO;2-M](https://doi.org/10.1002/1097-0258(20001130)19:22%3C3127::AID-SIM784%3E3.0.CO;2-M).
- Deininger, Klaus, Songqing Jin, Hari K. Nagarajan, and Fang Xia. 2011. *Does Female Reservation Affect Long-Term Political Outcomes? Evidence from Rural India*. No. 5708. World Bank. <https://document.s1.worldbank.org/curated/en/595231468040603055/pdf/WPS5708.pdf>.
- Duflo, Esther, and Petia Topalova. 2004. "Unappreciated Service: Performance, Perceptions, and Women Leaders in India." <https://poverty-action.org/sites/default/files/publications/unappreciated.pdf>.
- Gajwani, Kiran, and Xiaobo Zhang. 2015. "Gender and Public Goods Provision in Tamil Nadu's Village Governments." *The World Bank Economic Review* 29 (2): 234–61. <https://doi.org/10.1093/wber/lhu001>.
- Goyal, Tanushree. 2023. *Replication Data for: Local Political Representation as a Pathway to Power: A Natural Experiment in India*. Harvard Dataverse. <https://doi.org/10.7910/DVN/9XPV4I>.
- Green, Donald P., Manu Singh, and Gaurav Sood. 2026. "New Evidence on the Allocative Effects of Randomly Assigned Reservations for Women Leaders in Rural Indian Local Government." https://github.com/in-rolls/quota_spending.
- Heinze, Alyssa R., Rachel E. Brulé, and Simon Chauchard. 2025. "Who Actually Governs? Gender Inequality and Political Representation in Rural India." *Journal of Politics* 87 (2): 818–22. <https://doi.org/10.1086/730746>.
- Huidobro, Alba, Soledad Artiz Prillaman, and Deepak Singhanian. n.d. *Family Politics: Collective Governance and Women's Political Representation in Rural India*. Public research abstract on Soledad Artiz Prillaman's website. <https://www.soledadprillaman.com/research>.
- Iyer, Lakshmi, and Anandi Mani. 2019. "The Road Not Taken: Gender Gaps Along Paths to Political Power." *World Development* 119: 68–80. <https://doi.org/10.1016/j.worlddev.2019.03.004>.
- Jain, Chandan, Shagun Kashyap, Rahul Lahoti, and Soham Sahoo. 2022. *Do Educated Leaders Affect Economic Development? Evidence from India*. No. 15278. IZA. <https://docs.iza.org/dp15278.pdf>.
- Joshi, Gaurav, Amit Joshi, and Abhishek Kumar Pant. 2025. "Does Education Make Local Elected Officials Perform Better? A Case of Gram Panchayats in India." *Indian Journal of Public Administration* 71 (1): 53–61. <https://doi.org/10.1177/00195561241284449>.
- Karekurve-Ramachandra, Varun, and Alexander Lee. 2020a. "Can Gender Quotas Improve Public Service Provision? Evidence from Indian Local Government." https://www.rochester.edu/college/faculty/alexander_lee/wp-content/uploads/2020/12/gender-performance7.pdf.
- Karekurve-Ramachandra, Varun, and Alexander Lee. 2020b. *Replication Data for: Do Gender Quotas Hurt Less Privileged Groups? Evidence from India*. Harvard Dataverse. <https://doi.org/10.7910/DVN/0QOVCH>.
- Karekurve-Ramachandra, Varun, and Alexander Lee. 2023. *Can Gender Quotas Improve Public Service Provision? Evidence from Indian Local Government*. <https://doi.org/10.7910/DVN/IO9SLQ>.

- Karekurve-Ramachandra, Varun, and Gaurav Sood. 2026. “The Limits of Electoral Gender Quotas in Rural Local Bodies.” https://gsood.com/research/papers/quota_raj.pdf.
- Mori, Yuko, R. Manjula, D. Rajasekhar, Jun Goto, and Takashi Kurosaki. 2025. “Who Becomes a Politician in a Gender Quota System? Evidence from India.” *World Development* 188: 106911. <https://doi.org/10.1016/j.worlddev.2024.106911>.
- Prakash, Nishith, Marc Rockmore, and Yogesh Uppal. 2019. “Do Criminally Accused Politicians Affect Economic Outcomes? Evidence from India.” *Journal of Development Economics* 141: 102370. <https://doi.org/10.1016/j.jdeveco.2019.102370>.
- Priebe, Jan. 2017. “Political Reservation and Female Empowerment: Evidence from Maharashtra, India.” *Oxford Development Studies* 45 (4): 499–521. <https://doi.org/10.1080/13600818.2017.1298740>.
- Sood, Gaurav. 2022. *Local-Elections: Indian Local Election Data*. GitHub repository in-rolls/local_elections. https://github.com/in-rolls/local_elections/tree/5f2261a57435c6029a15697b96432012ee20b5de/data/delhi.
- State Election Commission, NCT of Delhi. 2012. *Notification Regarding Reservation of Seats, 2012*. https://sec.delhi.gov.in/sites/default/files/SEC/circulars-orders/notification-regarding-reservation-seats-2012_1.pdf.
- State Election Commission, NCT of Delhi. 2022. *Reservation Order for the Municipal Corporation of Delhi*. https://sec.delhi.gov.in/sites/default/files/SEC/generic_multiple_files/reservationorder_0.pdf.
- Vaidyanathan, R. 2023. *MCD Election Results 2022*. OpenCity. <https://data.opencity.in/dataset/delhi-mcd-elections-2022>.

A Measurement and reproducibility

A.1 Affidavit verification

We do not use the education recorded in the older Delhi files, whose coding also appears in Karekurve-Ramachandra and Lee’s deposit (Sood 2022; Karekurve-Ramachandra and Lee 2020b). For the 2012 winners, the deposited graduate classification agrees with the winners’ own MyNeta profiles 54% of the time, close to the agreement obtained after randomly shuffling the deposited values across winners (53%), while the same records’ declared assets match 97% of the time. The 2017 education labels agree with the profiles for 261 of 262 winners. In the Mumbai audit sample, the deposited education and pending-case indicators agree with MyNeta.

A.2 Asset measurement

The asset regressions use $\log(A + 1)$, where A is declared assets in rupees. The added rupee retains zero declarations. Percentage differences are $100[\exp(\hat{\beta}) - 1]$, with the same transformation applied to the confidence limits.

A.3 Schooling synthesis

For years of schooling, we divide the reported difference and its standard error by the pooled within-group SD. For binary thresholds, $d = \log(OR)\sqrt{3}/\pi$, and the variance is multiplied by $3/\pi^2$. This is the logistic-distribution conversion in `metafor`’s `OR2DL` measure (Chinn 2000). Published threshold comparisons use the reported group shares and sizes, with counts rounded to integers. Our comparisons use the reservation coefficient from a fixed-effects logistic regression, with the same controls and clustering as the percentage-point models. Their individual intervals use t_{G-1} critical values. Fixed-effect groups with no outcome variation do not identify the logistic coefficient and are excluded; Table A.1 reports the resulting coverage.

Table A.1: Samples used to standardize this paper’s schooling estimates.

Election	Classified winners	Logit sample	Clusters
Bihar 2016	7,873	7,180	456
Uttar Pradesh 2010	51,566	51,566	72
Uttar Pradesh 2015	59,019	57,644	803
Uttar Pradesh 2021	47,228	47,012	717
Rajasthan 2020	7,748	7,671	345
Kerala 2010	15,024	15,024	14
Kerala 2015	14,121	14,121	14
Kerala 2020	13,457	13,457	14
Mumbai 2012 and 2017	449	449	227
Delhi 2012	226	171	47
Delhi 2017	262	203	50
Delhi 2022	244	197	53

The combined model is $d_{sj} = \mu + u_s + v_{sj} + e_{sj}$, where s indexes a published sample or one of our six settings, j indexes an estimate, and e_{sj} has the estimated sampling variance. We estimate both random-effect variances by restricted maximum likelihood and use t intervals with containment

degrees of freedom. The main model assumes independent sampling errors conditional on the random effects; a sensitivity model sets their correlation to 0.5 within settings. The prediction interval includes both random-effect variances. Separate group means distinguish published village heads, our northern village heads, and our Kerala ward members and city councillors.

Chattopadhyay and Duflo represent Birbhum in the main pool; Beaman et al. replace them in a sensitivity check. Bamezai et al.’s Table 7, Panel B, column 1 estimates a fuzzy-RD effect near reservation thresholds on winners’ education standardized against GP citizens. Column 2 uses a same-gender citizen benchmark; column 3 substitutes husbands’ education. None enters our pool: the first has a different target population and standardization, and the others also change the comparison or whose qualifications are measured. Their 2016 Bihar sample overlaps ours; our Bihar estimate is the only one used in the pool. Earlier Delhi and Mumbai deposit-based estimates are not additional independent evidence. Comparisons of women and men, candidate outcomes, and selection relative to citizens are excluded. The pool is drawn from the studies we located and could extract, not a systematic census of the literature. Reconstructed standard errors for published threshold comparisons omit clustering and repeated village observations.

Table A.2: Combined schooling synthesis: alternative samples and dependence assumptions.

Specification	Estimates	Difference [95% CI], SD
Combined	16	-0.45 [-0.76, -0.14]
Alternative Birbhum study	16	-0.43 [-0.72, -0.13]
Without Rajasthan	15	-0.48 [-0.82, -0.13]
Error correlation: 0.5	16	-0.45 [-0.76, -0.15]

A.3.1 Additional schooling comparisons from Bhavnani et al.

Table 2 of Bhavnani et al. (2024) reports reservation-category coefficients against fully open seats. Within SC/ST-reserved seats, the women’s-reservation difference is the SC/ST-women coefficient minus the SC/ST-open coefficient. Table A.3 reconstructs these comparisons. The coefficients’ covariance is unreported, so the standard error of their difference can range from the absolute difference to the sum of their standard errors. The displayed interval uses that largest standard error and a normal approximation; it is an uncertainty envelope, not an exact reconstructed confidence interval.

Table A.3: Women’s-reservation schooling differences in years, reconstructed from Bhavnani et al. (2024), Table 2, columns 4 and 6.

Sample	Caste	Difference	SE range	95% envelope
Odisha 2022	Non-SC/ST	-1.08	0.39	[-1.84, -0.32]
Odisha 2022	SC/ST	-2.29	0.12 to 0.74	[-3.74, -0.84]
REDS 2014-16	Non-SC/ST	-1.55	0.92	[-3.35, 0.25]
REDS 2014-16	SC/ST	-0.58	0.57 to 2.71	[-5.89, 4.73]

Odisha uses the 2022 reservation schedule and schooling from the 2011–12 census for matched winners. The other sample comes from the REDS listing survey of 2014–16; the draft does not specify a common election year. The reconstructed differences retain the source’s covariate adjustment

and matching restrictions. The reported tables lack the schooling SDs needed for standardization and the design information needed to recover our overall caste-adjusted reservation coefficient. We therefore report these results in years and keep them outside the pooled synthesis.

A.4 Coverage and missing outcomes

Table A.4: Education coverage among eligible winner records. N includes unclassified qualifications.

State	Year	Office	Seat	Classified (%)	N
UP	2010	Village head	Open	99.6	34,654
UP	2010	Village head	Women	99.4	17,166
UP	2015	Village head	Open	100.0	39,093
UP	2015	Village head	Women	100.0	19,926
UP	2021	Village head	Open	95.3	32,997
UP	2021	Village head	Women	94.1	16,774
Rajasthan	2020	Village head	Open	81.8	5,695
Rajasthan	2020	Village head	Women	56.9	5,433
Kerala	2010	Block member	Open	91.0	983
Kerala	2010	Block member	Women	93.0	1,095
Kerala	2010	Ward member	Open	89.7	8,047
Kerala	2010	Ward member	Women	90.8	8,594
Kerala	2010	District member	Open	87.7	162
Kerala	2010	District member	Women	94.7	169
Kerala	2015	Block member	Open	87.4	1,021
Kerala	2015	Block member	Women	88.7	1,041
Kerala	2015	Ward member	Open	87.6	7,725
Kerala	2015	Ward member	Women	89.7	8,196
Kerala	2015	District member	Open	86.0	164
Kerala	2015	District member	Women	92.2	167
Kerala	2020	Block member	Open	88.7	980
Kerala	2020	Block member	Women	89.3	1,098
Kerala	2020	Ward member	Open	83.8	7,726
Kerala	2020	Ward member	Women	84.9	8,226
Kerala	2020	District member	Open	86.1	166
Kerala	2020	District member	Women	84.2	165

Education is missing for many winners in Rajasthan, where qualifications are often recorded as “Other”, and is classified at different rates across Kerala’s seat types. Ages outside 21 to 100 are set to missing. Some Uttar Pradesh inputs come from integration work not yet in a published release of the election data; the repository keeps their byte hashes and the adapter diff.

The missing-education bounds keep the all-winner regression design and let each unclassified winner be either a graduate or not. After reservation is residualized on the controls, the coefficient is a weighted sum of outcomes, so assigning each missing outcome by the sign of its weight gives the exact minimum and maximum. The bounds carry no sampling uncertainty. Uttar Pradesh’s bounds are negative in each election; Rajasthan’s and Kerala’s include both signs.

Table A.5: Missing-education bounds on the adjusted all-winner graduate-share difference (percentage points). These are not confidence intervals.

State	Year	Office	Lower	Upper	Missing	N
UP	2010	Village head	-13.5	-12.4	254	51,820
UP	2015	Village head	-9.1	-9.1	0	59,019
UP	2021	Village head	-12.0	-1.3	2,543	49,771
Rajasthan	2020	Village head	-29.7	31.5	3,380	11,128
Kerala	2010	Block member	-12.3	3.4	165	2,078
Kerala	2010	Ward member	-10.6	8.9	1,617	16,641
Kerala	2010	District member	-12.7	4.8	29	331
Kerala	2015	Block member	-13.8	10.1	247	2,062
Kerala	2015	Ward member	-12.2	10.5	1,800	15,921
Kerala	2015	District member	-18.0	3.7	36	331
Kerala	2020	Block member	-14.5	7.5	228	2,078
Kerala	2020	Ward member	-14.4	16.9	2,495	15,952
Kerala	2020	District member	-15.1	15.2	49	331

B Complete rural estimates

B.1 Bihar: education

Table B.1: Bihar rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2016	Village head	-15.0 [-16.5, -13.4]	1.5 [0.9, 2.1]	7,873	523	Block
2016	Ward member	-4.6 [-4.9, -4.2]	11.5 [10.8, 12.3]	97,200	517	Block
2016	Sarpanch	-12.1 [-13.4, -10.8]	4.7 [3.7, 5.6]	7,841	521	Block
2016	Panch	-2.8 [-3.1, -2.6]	14.8 [14.0, 15.7]	85,355	516	Block
2016	Block member	-10.8 [-12.0, -9.5]	3.2 [2.5, 4.0]	10,743	519	Block
2016	District member	-12.7 [-18.4, -6.9]	1.3 [-0.4, 2.9]	888	38	District

B.2 Uttar Pradesh: education

Table B.2: Uttar Pradesh rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2010	Village head	-13.1 [-14.5, -11.7]	6.3 [5.1, 7.4]	51,566	72	District
2015	Village head	-9.1 [-9.7, -8.5]	9.3 [8.7, 10.0]	59,019	823	Block
2021	Village head	-7.4 [-8.1, -6.7]	8.2 [7.5, 9.0]	47,228	728	Block

B.3 Rajasthan: education

Table B.3: Rajasthan rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2020	Village head	-6.9 [-9.0, -4.9]	0.1 [-0.1, 0.2]	7,748	352	Block

B.4 Kerala: education

Table B.4: Kerala rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included. Illiteracy is unavailable.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2010	Ward member	-0.5 [-2.4, 1.3]	—	15,024	14	District
2010	Block member	-4.1 [-7.9, -0.3]	—	1,913	150	Block
2010	District member	-4.3 [-14.1, 5.6]	—	302	14	District
2015	Ward member	-0.1 [-1.6, 1.3]	—	14,121	14	District
2015	Block member	-1.2 [-5.0, 2.6]	—	1,815	148	Block
2015	District member	-7.8 [-21.1, 5.4]	—	295	14	District
2020	Ward member	2.0 [-0.8, 4.7]	—	13,457	14	District
2020	Block member	-3.5 [-7.5, 0.4]	—	1,850	149	Block
2020	District member	-1.8 [-15.6, 12.1]	—	282	14	District

B.5 Age

Table B.5: Age of rural officials: differences in years with pointwise 95% confidence intervals. N: winners with age recorded; G: geographic clusters. Controls match the education models.

State	Year	Office	Difference [95% CI]	N	G
Bihar	2016	Village head	-3.6 [-4.1, -3.2]	7,934	524
Bihar	2016	Ward member	-2.0 [-2.2, -1.8]	97,211	517
Bihar	2016	Sarpanch	-5.6 [-6.2, -5.1]	7,871	521
Bihar	2016	Panch	-2.9 [-3.1, -2.7]	85,376	516
Bihar	2016	Block member	-2.7 [-3.1, -2.3]	10,775	519
Bihar	2016	District member	-4.2 [-5.6, -2.7]	891	38
Bihar	2021	Village head	-3.2 [-3.6, -2.7]	8,050	533
Bihar	2021	Ward member	-0.5 [-0.6, -0.3]	109,501	533
Bihar	2021	Sarpanch	-6.2 [-6.7, -5.7]	8,043	533
Bihar	2021	Panch	-3.2 [-3.4, -3.0]	106,636	533
Bihar	2021	Block member	-2.3 [-2.7, -1.9]	10,972	533
Bihar	2021	District member	-3.9 [-5.0, -2.8]	1,144	38
Uttar Pradesh	2010	Village head	-1.4 [-1.6, -1.2]	51,662	72
Uttar Pradesh	2021	Village head	-1.3 [-1.5, -1.1]	49,757	728

State	Year	Office	Difference [95% CI]	N	G
Rajasthan	2020	Village head	-1.7 [-2.3, -1.1]	11,113	352
Kerala	2010	Ward member	-4.8 [-5.9, -3.8]	14,793	14
Kerala	2010	Block member	-4.9 [-5.8, -4.0]	1,980	151
Kerala	2010	District member	-4.2 [-7.3, -1.2]	330	14
Kerala	2015	Ward member	-4.2 [-5.2, -3.2]	15,724	14
Kerala	2015	Block member	-3.9 [-4.9, -3.0]	2,029	151
Kerala	2015	District member	-3.0 [-5.8, -0.3]	327	14
Kerala	2020	Ward member	-3.5 [-4.5, -2.4]	15,860	14
Kerala	2020	Block member	-3.1 [-4.0, -2.2]	2,070	152
Kerala	2020	District member	-3.6 [-6.0, -1.2]	331	14

B.6 Occupation, assets and criminal records

Table B.6: Occupation, assets and criminal records: reserved minus open seats, with the controls and clustering of the education models. No occupation and criminal record in percentage points; log declared assets in log points. No occupation counts nil, unemployed, homemaker and student entries. Mumbai, which records no occupation, reports whether the councillor declared a tax ID (PAN).

Outcome	State	Year	Office	Difference [95% CI]	N
No occupation	Delhi	2012	Councillor	28.0 [18.4, 37.5]	231
No occupation	Delhi	2017	Councillor	25.6 [17.1, 34.0]	264
No occupation	Delhi	2022	Councillor	21.7 [13.5, 29.9]	246
No occupation	Kerala	2010	Block member	38.8 [33.6, 43.9]	1,690
No occupation	Kerala	2010	Ward member	54.5 [50.2, 58.8]	13,992
No occupation	Kerala	2010	District member	12.4 [0.9, 23.8]	241
No occupation	Kerala	2015	Block member	35.3 [30.4, 40.3]	1,654
No occupation	Kerala	2015	Ward member	50.2 [44.9, 55.4]	12,647
No occupation	Kerala	2015	District member	15.1 [6.7, 23.6]	240
No occupation	Kerala	2020	Block member	30.7 [25.5, 35.9]	1,637
No occupation	Kerala	2020	Ward member	41.0 [35.7, 46.2]	12,267
No occupation	Kerala	2020	District member	0.2 [-10.3, 10.7]	217
No occupation	Rajasthan	2020	Village head	51.1 [48.2, 53.9]	11,128
No tax ID declared	Mumbai	2012, 2017	Councillor	5.8 [2.1, 9.4]	449
Log declared assets	Rajasthan	2020	Village head	-0.64 [-0.77, -0.51]	11,128
Log declared assets	Uttar Pradesh	2021	Village head	-0.10 [-0.14, -0.05]	49,771
Criminal record	Uttar Pradesh	2021	Village head	-0.04 [-0.12, 0.04]	42,720

C Delhi estimates and coverage

Table C.1: Delhi councillors: adjusted differences in percentage points, except age in years. Assembly-constituency and caste-reservation controls; pointwise intervals clustered by assembly constituency. G denotes clusters. The occupation sensitivity excludes social workers.

Year	Outcome	Difference [95% CI]	N	G
2012	Graduate or above	-0.6 [-14.4, 13.2]	226	65
2012	Illiterate	2.5 [-1.3, 6.3]	226	65
2012	Age (years)	-6.0 [-9.0, -3.0]	235	65
2012	Any pending case	-23.5 [-35.2, -11.8]	237	65
2012	No occupation	28.0 [18.4, 37.5]	231	64
2012	No occupation, sensitivity	30.9 [20.7, 41.1]	213	62
2017	Graduate or above	1.1 [-10.4, 12.6]	262	68
2017	Illiterate	0.6 [-1.7, 2.8]	262	68
2017	Age (years)	-3.6 [-5.6, -1.5]	262	68
2017	Any pending case	-10.6 [-18.4, -2.8]	267	68
2017	No occupation	25.6 [17.1, 34.0]	264	68
2017	No occupation, sensitivity	27.0 [18.1, 35.8]	245	68
2022	Graduate or above	-3.4 [-15.9, 9.1]	244	68
2022	Illiterate	2.4 [-1.0, 5.7]	244	68
2022	Age (years)	-2.3 [-4.8, 0.3]	248	68
2022	Any pending case	-21.0 [-29.9, -12.0]	248	68
2022	No occupation	21.7 [13.5, 29.9]	246	68
2022	No occupation, sensitivity	24.7 [15.5, 33.9]	226	68

Table C.2: Delhi outcome coverage among recorded winners. Percentages use all winners of each seat type as the denominator.

Year	Seat	N	Education (%)	Age (%)	Cases (%)
2012	Open	134	85.8	86.6	87.3
2012	Women	138	80.4	86.2	87.0
2017	Open	134	97.0	98.5	99.3
2017	Women	138	95.7	94.2	97.1
2022	Open	125	97.6	100.0	100.0
2022	Women	125	97.6	98.4	98.4

Table C.3: Delhi missing-outcome bounds on the adjusted all-winner coefficient, in percentage points. They carry no sampling uncertainty.

Year	Outcome	Lower	Upper	Missing	N
2012	Graduate or above	-16.4	18.9	46	272
2012	Any pending case	-33.9	-6.1	35	272
2017	Graduate or above	-2.5	6.3	10	272
2017	Any pending case	-11.5	-7.5	5	272
2022	Graduate or above	-5.0	-0.1	6	250
2022	Any pending case	-21.3	-19.9	2	250

D Published comparisons

Table D.1: Published comparisons of leader qualifications

Study	Setting	Design	Outcome	Reserved	Open	Difference (SE)	N
<i>Elected village leaders, women-reserved vs. open seats</i>							
Chattopadhyay and Duflo (2001) <i>Table 3, panels A and B</i>	West Bengal (Birbhum); 1998 term; GP pradhan	Randomized rotation	Years of schooling (years)	7.13	9.92	−2.79 (0.54)	161
			Literate (%)	80.0	98.0	−19.0 (4.0)	161
			Elected to GP council before 1998 (%)	11.0	43.0	−32.0 (7.0)	161
Beaman et al. (2009) <i>Table II, column 6, printed p. 1507</i>	West Bengal (Birbhum); 2003 term; GP pradhan	Randomized rotation; adjusted	Years of schooling (years)			−2.104 (0.548)	165
			First time as pradhan in 2003 term (%)			26.5 (6.8)	165
			Other political experience (%)			−12.1 (7.9)	165
Ban and Rao (2007) <i>Table 8 (means), Table 9 column 2 (difference)</i>	Andhra Pradesh, Karnataka, Kerala, Tamil Nadu; 2002 survey; GP president	Rotation; caste-reserved GPs excluded; adjusted	Years of schooling (years)	6.074	8.456	−2.620 (0.766)	106
			Political knowledge score	3.185	4.608	−1.413 (0.292)	106
			Prior political positions (count)			−0.461 (0.188)	106
Deininger et al. (2011) <i>Table 3, printed p. 22</i>	National (REDS, 233 villages); Up to three elections before REDS 2007; GP pradhan	Descriptive	Illiterate (%)	24.7	5.7	19.0 (3.4) [†]	639
			At least secondary education (%)	28.6	62.7	−34.0 (4.1) [†]	639
			Held political office before (%)	11.7	24.2	−12.6 (3.1) [†]	639
Afridi, Iversen and Sharan (2013) <i>Table 2, panel I, printed p. 38</i>	Andhra Pradesh (8 districts); In office 2006–10; GP sarpanch	Descriptive	Illiterate (%)	9.0	0.0	9.0 (2.2)	297
			Higher secondary or more (%)	15.0	54.0	−40.0 (5.2)	297
			Own prior political experience (%)	11.0	22.0	−11.0 (4.4)	297
Bamezai et al. (2026) <i>Table 7, panel B</i>	Bihar; 2016 and 2021 elections; Mukhiya	Fuzzy RD at population threshold; adjusted	Schooling, SD of all citizens			−0.921 (0.073)	3,294
<i>Elected leaders, reserved women vs. men in other seats</i>							
Raabe, Sekher and Birner (2009) <i>Table 4</i>	Karnataka (12 districts); 2006 survey; GP president	Descriptive	Illiterate (%)	9.0	4.0	5.0 (6.6) [†]	71

Table D.1 (continued)

Study	Setting	Design	Outcome	Reserved	Open	Difference (SE)	N
			Degree or above (%)	0.0	19.0	−19.0 (5.7) [†]	71
<i>Elected village leaders, women vs. men (conditional comparison)</i>							
Gajwani and Zhang (2015) <i>Table 5, column 6, p. 248</i>	Tamil Nadu (144 GPs); 2005; GP president	OLS; adjusted	Panchayat knowledge (0–19)			−3.199 (0.474)	141
<i>Municipal councillors, women-reserved vs. open seats</i>							
Karekurve-Ramachandra and Lee (2020) <i>Table 1, printed p. 14</i>	Mumbai; 2011–2018 (2012 and 2017 terms); Municipal councillor	Descriptive	Attended university (%)	43.7	39.0	4.7 (2.9) [†]	1,118
			Criminal record (%)	3.7	30.3	−26.6 (2.1) [†]	1,118
<i>Candidates, women-reserved vs. open seats</i>							
Mori et al. (2022) <i>Table 3, panel B, column 1</i>	Karnataka (Kolar, Tumkur); December 2020 election; GP member (single-seat)	Random rotation; adjusted	Education position among all villagers			−0.41 (0.33)	97
Bamezai et al. (2026) <i>Table 7, panel A</i>	Bihar; 2016 and 2021 elections; Mukhiya	Fuzzy RD at population threshold; adjusted	Schooling, SD of all citizens			−0.765 (0.024)	44,583
<i>Elected leaders, other benchmarks</i>							
Mori et al. (2022) <i>Table 3, panel B, column 3</i>	Karnataka (Kolar, Tumkur); December 2020 election; GP member (single-seat)	Random rotation; adjusted	Education position among same-gender villagers <i>Reserved winner's position among women vs. open winner's position among men</i>			0.74 (0.44)	92
Bamezai et al. (2026) <i>Table 7, panel B</i>	Bihar; 2016 and 2021 elections; Mukhiya	Fuzzy RD at population threshold; adjusted	Schooling, SD of same-gender citizens <i>Reserved winner's position among women vs. open winner's position among men</i>			−0.619 (0.074)	3,249
			Husband's schooling, SD of all citizens <i>Reserved winner's husband vs. open-seat winner</i>			−0.364 (0.089)	3,058

Reserved and Open are group means; shares are percentages. Differences are reserved minus open, in outcome units (percentage points for shares), with standard errors in parentheses. Adjusted differences are regression coefficients. A dagger marks an SE reconstructed from shares and sample sizes or a reported t-statistic; these ignore clustering and repeated units. N is the combined group size or regression sample. SD rows use the GP's citizen distribution; selection scores measure position in the villager education distribution. Chattopadhyay and Duflo (2001) and Beaman et al. (2009) study the same Birbhum GPs in successive terms; the published version of Chattopadhyay and Duflo (2004) omits the pradhan-characteristics table. Karekurve-Ramachandra and Lee (2020) use the same Mumbai councillor data as this paper. Mori et al.'s published version (2025) reports the same-gender benchmark. Row-level source locations and transcription notes are in the repository's evidence/literature/studies files. Gajwani and Zhang's adjusted female-minus-male knowledge coefficient is excluded from the schooling meta-analysis; group means are unavailable. Bhavnani et al. (2024) contrasts appear in Table A.3 and are not pooled.